

1. **Permissioned Network (Phase 1):**

- Trusted nodes (oracles) for performance verification

2. **Transition to Permissionless Network (Phase 2):**

- All nodes participate in performance measurement

Key Transition:

- Phase 1: Trusted nodes (oracles) verify performance
- Phase 2: Every node pings and reports on every other node operator

In Phase 1, the oracle node plays a crucial role within the Network. It functions as both a data collector and a regular service operator. As a data collector, the oracle node gathers performance metrics from all active and reserve nodes in the Network. This phased approach allows for a controlled and secure evolution of the Network, balancing the need for initial stability with the long-term goal of decentralization.

5.1.4 Service Node Slashing

To maintain a high-performance network, the Network incorporates a slashing mechanism that penalizes underperforming nodes. This mechanism is directly tied to the previously defined node performance coefficient $U(i, t)$.

Slashing Threshold: The governance will define a performance threshold τ , where $0 < \tau < 1$. If a node's performance coefficient falls below this threshold, it triggers a slashing event.

The slashing process is as follows:

1. **Performance Evaluation:** The performance coefficient $U(i, t)$ is continuously calculated for each node i at regular intervals t .
2. **Threshold Check:** If $U(i, t) < \tau$, a slashing event is triggered for node i .
3. **Jailing:** Upon a slashing event, the underperforming node is immediately moved to the "jailed" set. This set contains nodes that are temporarily or permanently removed from active participation in the Network.
4. **Replacement:** Once a node is jailed, it is replaced in the active set by the next eligible node. This enables the Network to maintain its designated number of active nodes at all times.
5. **Stake Reduction:** The jailed node may face a reduction in its staked tokens as a penalty for underperformance. The exact percentage of stake reduction will be determined by Network governance.